

Полная таблица тестов НСС C89→C51

Источник: artifacts/test-matrix.json · сгенерировано: `` · строк: 107 · перенос ячеек: 56 симв.

Колонки **Вход** / **Ожидание** / **Выход** — факты прогона Haskell (TestMatrix).

№	Вход	Ожидание	Выход
1	int main() { return 0; }	int main() { return 0; }	int main() { return 0; }
2	int x; void f() {}	[TokenInt,TokenIdentifier "x",TokenSemicolon,TokenVoid, TokenIdentifier "f",TokenLeftParen,TokenRightParen, TokenLeftBrace,TokenRightBrace]	[TokenInt,TokenIdentifier "x",TokenSemicolon,TokenVoid, TokenIdentifier "f",TokenLeftParen,TokenRightParen, TokenLeftBrace,TokenRightBrace]
3	int main() { return 0; }	[TokenInt,TokenIdentifier "main",TokenLeftParen, TokenRightParen,TokenLeftBrace,TokenReturn,TokenNumber 0, TokenSemicolon,TokenRightBrace]	[TokenInt,TokenIdentifier "main",TokenLeftParen, TokenRightParen,TokenLeftBrace,TokenReturn,TokenNumber 0, TokenSemicolon,TokenRightBrace]
4	a==b; a!=b; a+=1; a- =2; a+b-c*d;	[TokenIdentifier "a",TokenEqual,TokenIdentifier "b", TokenSemicolon,TokenIdentifier "a",TokenBangEqual, TokenIdentifier "b",TokenSemicolon,TokenIdentifier "a", TokenPlusAssign,TokenNumber 1,TokenSemicolon, TokenIdentifier "a",TokenMinusAssign,TokenNumber 2, TokenSemicolon,TokenIdentifier "a",TokenPlus, TokenIdentifier "b",TokenMinus,TokenIdentifier "c", TokenMultiply,TokenIdentifier "d",TokenSemicolon]	[TokenIdentifier "a",TokenEqual,TokenIdentifier "b", TokenSemicolon,TokenIdentifier "a",TokenBangEqual, TokenIdentifier "b",TokenSemicolon,TokenIdentifier "a", TokenPlusAssign,TokenNumber 1,TokenSemicolon, TokenIdentifier "a",TokenMinusAssign,TokenNumber 2, TokenSemicolon,TokenIdentifier "a",TokenPlus, TokenIdentifier "b",TokenMinus,TokenIdentifier "c", TokenMultiply,TokenIdentifier "d",TokenSemicolon]
5	a	[TokenIdentifier "a"]	[TokenIdentifier "a"]
6	42	[TokenNumber 42]	[TokenNumber 42]
7	0 077 0x1f 0X2A	[TokenNumber 0,TokenNumber 63,TokenNumber 31, TokenNumber 42]	[TokenNumber 0,TokenNumber 63,TokenNumber 31, TokenNumber 42]
8	10	[TokenNumber 8]	[TokenNumber 8]
9	8	[TokenLexError "Invalid octal literal: 08"]	[TokenLexError "Invalid octal literal: 08"]
10	0x	[TokenLexError "Invalid hexadecimal literal: 0x"]	[TokenLexError "Invalid hexadecimal literal: 0x"]
11	10U 10l 10UL 0xFFu 077L	[TokenNumberWithSuffix 10 SufU, TokenNumberWithSuffix 10 SufL, TokenNumberWithSuffix 10 SufUL, TokenNumberWithSuffix 255 SufU, TokenNumberWithSuffix 63 SufL]	[TokenNumberWithSuffix 10 SufU, TokenNumberWithSuffix 10 SufL, TokenNumberWithSuffix 10 SufUL, TokenNumberWithSuffix 255 SufU, TokenNumberWithSuffix 63 SufL]
12	0 00 075 09 0x1F 0X1f 123u 077L	[TokenNumber 0,TokenNumber 0,TokenNumber 61, TokenLexError "Invalid octal literal: 09",TokenNumber 31,	[TokenNumber 0,TokenNumber 0,TokenNumber 61, TokenLexError "Invalid octal literal: 09",TokenNumber 31,

[illegible]

		TokenSigned,TokenVoid,TokenDefault,TokenGoto,TokenSizeof, TokenVolatile,TokenDo,TokenIf,TokenStatic,TokenWhile, TokenSfr,TokenSfr16,TokenSbit,TokenSft,TokenBit, TokenData,TokenIdata,TokenBdata,TokenPdata,TokenXdata, TokenCode,TokenInterrupt,TokenUsing,TokenReentrant, TokenAt,TokenIdentifier "ident",TokenNumber 42, TokenAssign,TokenEqual,TokenBangEqual,TokenPlus, TokenPlusAssign,TokenPlusPlus,TokenMinus, TokenMinusAssign,TokenMinusMinus,TokenMultiply, TokenMultiplyEqual,TokenDivide,TokenDivideEqual, TokenSemicolon,TokenComma,TokenDot,TokenColon, TokenQuestion,TokenBang,TokenHash,TokenPercent, TokenPercentEqual,TokenAmpersand,TokenAmpersandAmpersand, TokenAmpersandEqual,TokenPipe,TokenPipePipe, TokenPipeEqual,TokenTilde,TokenCaret,TokenCaretEqual, TokenBackslash,TokenStringLiteral "s", TokenCharLiteral 'x',TokenLeftParen,TokenRightParen, TokenLeftBrace,TokenRightBrace,TokenLeftBracket, TokenRightBracket,TokenLeftAngle,TokenLessEqual, TokenLessLess,TokenLessLessEqual,TokenRightAngle, TokenGreaterEqual,TokenGreaterGreater, TokenGreaterGreaterEqual]	TokenSigned,TokenVoid,TokenDefault,TokenGoto,TokenSizeof, TokenVolatile,TokenDo,TokenIf,TokenStatic,TokenWhile, TokenSfr,TokenSfr16,TokenSbit,TokenSft,TokenBit, TokenData,TokenIdata,TokenBdata,TokenPdata,TokenXdata, TokenCode,TokenInterrupt,TokenUsing,TokenReentrant, TokenAt,TokenIdentifier "ident",TokenNumber 42, TokenAssign,TokenEqual,TokenBangEqual,TokenPlus, TokenPlusAssign,TokenPlusPlus,TokenMinus, TokenMinusAssign,TokenMinusMinus,TokenMultiply, TokenMultiplyEqual,TokenDivide,TokenDivideEqual, TokenSemicolon,TokenComma,TokenDot,TokenColon, TokenQuestion,TokenBang,TokenHash,TokenPercent, TokenPercentEqual,TokenAmpersand,TokenAmpersandAmpersand, TokenAmpersandEqual,TokenPipe,TokenPipePipe, TokenPipeEqual,TokenTilde,TokenCaret,TokenCaretEqual, TokenBackslash,TokenStringLiteral "s", TokenCharLiteral 'x',TokenLeftParen,TokenRightParen, TokenLeftBrace,TokenRightBrace,TokenLeftBracket, TokenRightBracket,TokenLeftAngle,TokenLessEqual, TokenLessLess,TokenLessLessEqual,TokenRightAngle, TokenGreaterEqual,TokenGreaterGreater, TokenGreaterGreaterEqual]
22	a	[TokenIdentifier "a"]	[TokenIdentifier "a"]
23	42	[TokenNumber 42]	[TokenNumber 42]
24	a	[TokenIdentifier "a"]	[TokenIdentifier "a"]
25	b	[TokenIdentifier "b"]	[TokenIdentifier "b"]
26	ab	[TokenIdentifier "ab"]	[TokenIdentifier "ab"]
27	"ok" 'a'	[TokenStringLiteral "ok",TokenCharLiteral 'a']	[TokenStringLiteral "ok",TokenCharLiteral 'a']
28	int main() { return 0; }	[TokenInt,TokenIdentifier "main",TokenLeftParen, TokenRightParen,TokenLeftBrace,TokenReturn,TokenNumber 0, TokenSemicolon,TokenRightBrace]	[TokenInt,TokenIdentifier "main",TokenLeftParen, TokenRightParen,TokenLeftBrace,TokenReturn,TokenNumber 0, TokenSemicolon,TokenRightBrace]
29	int main(void) { switch (x) { case 0: break; } return 0;	[TokenInt,TokenIdentifier "main",TokenLeftParen, TokenVoid,TokenRightParen,TokenLeftBrace,TokenSwitch, TokenLeftParen,TokenIdentifier "x",TokenRightParen, TokenLeftBrace,TokenCase,TokenNumber 0,TokenColon, TokenBreak,TokenSemicolon,TokenRightBrace,TokenReturn, TokenNumber 0,TokenSemicolon,TokenRightBrace]	[TokenInt,TokenIdentifier "main",TokenLeftParen, TokenVoid,TokenRightParen,TokenLeftBrace,TokenSwitch, TokenLeftParen,TokenIdentifier "x",TokenRightParen, TokenLeftBrace,TokenCase,TokenNumber 0,TokenColon, TokenBreak,TokenSemicolon,TokenRightBrace,TokenReturn, TokenNumber 0,TokenSemicolon,TokenRightBrace]

	}		
30	int main(void) { do { x = 1; } while (x); return 0; }	[TokenInt,TokenIdentifier "main",TokenLeftParen, TokenVoid,TokenRightParen,TokenLeftBrace,TokenDo, TokenLeftBrace,TokenIdentifier "x",TokenAssign, TokenNumber 1,TokenSemicolon,TokenRightBrace,TokenWhile, TokenLeftParen,TokenIdentifier "x",TokenRightParen, TokenSemicolon,TokenReturn,TokenNumber 0,TokenSemicolon, TokenRightBrace]	[TokenInt,TokenIdentifier "main",TokenLeftParen, TokenVoid,TokenRightParen,TokenLeftBrace,TokenDo, TokenLeftBrace,TokenIdentifier "x",TokenAssign, TokenNumber 1,TokenSemicolon,TokenRightBrace,TokenWhile, TokenLeftParen,TokenIdentifier "x",TokenRightParen, TokenSemicolon,TokenReturn,TokenNumber 0,TokenSemicolon, TokenRightBrace]
31	struct S { int x; }; int main(void) { return 0; }	[TokenStruct,TokenIdentifier "S",TokenLeftBrace,TokenInt, TokenIdentifier "x",TokenSemicolon,TokenRightBrace, TokenSemicolon,TokenInt,TokenIdentifier "main", TokenLeftParen,TokenVoid,TokenRightParen,TokenLeftBrace, TokenReturn,TokenNumber 0,TokenSemicolon, TokenRightBrace]	[TokenStruct,TokenIdentifier "S",TokenLeftBrace,TokenInt, TokenIdentifier "x",TokenSemicolon,TokenRightBrace, TokenSemicolon,TokenInt,TokenIdentifier "main", TokenLeftParen,TokenVoid,TokenRightParen,TokenLeftBrace, TokenReturn,TokenNumber 0,TokenSemicolon, TokenRightBrace]
32	int main(void) { switch (y) { default: break; } return 1; }	[TokenInt,TokenIdentifier "main",TokenLeftParen, TokenVoid,TokenRightParen,TokenLeftBrace,TokenSwitch, TokenLeftParen,TokenIdentifier "y",TokenRightParen, TokenLeftBrace,TokenDefault,TokenColon,TokenBreak, TokenSemicolon,TokenRightBrace,TokenReturn,TokenNumber 1, TokenSemicolon,TokenRightBrace]	[TokenInt,TokenIdentifier "main",TokenLeftParen, TokenVoid,TokenRightParen,TokenLeftBrace,TokenSwitch, TokenLeftParen,TokenIdentifier "y",TokenRightParen, TokenLeftBrace,TokenDefault,TokenColon,TokenBreak, TokenSemicolon,TokenRightBrace,TokenReturn,TokenNumber 1, TokenSemicolon,TokenRightBrace]
33	int main(void) { switch (v) { case 0: case 1: x = 2; break; } return 0; }	[TokenInt,TokenIdentifier "main",TokenLeftParen, TokenVoid,TokenRightParen,TokenLeftBrace,TokenSwitch, TokenLeftParen,TokenIdentifier "v",TokenRightParen, TokenLeftBrace,TokenCase,TokenNumber 0,TokenColon, TokenCase,TokenNumber 1,TokenColon,TokenIdentifier "x", TokenAssign,TokenNumber 2,TokenSemicolon,TokenBreak, TokenSemicolon,TokenRightBrace,TokenReturn,TokenNumber 0, TokenSemicolon,TokenRightBrace]	[TokenInt,TokenIdentifier "main",TokenLeftParen, TokenVoid,TokenRightParen,TokenLeftBrace,TokenSwitch, TokenLeftParen,TokenIdentifier "v",TokenRightParen, TokenLeftBrace,TokenCase,TokenNumber 0,TokenColon, TokenCase,TokenNumber 1,TokenColon,TokenIdentifier "x", TokenAssign,TokenNumber 2,TokenSemicolon,TokenBreak, TokenSemicolon,TokenRightBrace,TokenReturn,TokenNumber 0, TokenSemicolon,TokenRightBrace]
34	int main() { return 200; }	[TokenInt,TokenIdentifier "main",TokenLeftParen, TokenRightParen,TokenLeftBrace,TokenReturn, TokenNumber 200,TokenSemicolon,TokenRightBrace]	[TokenInt,TokenIdentifier "main",TokenLeftParen, TokenRightParen,TokenLeftBrace,TokenReturn, TokenNumber 200,TokenSemicolon,TokenRightBrace]
	sfr P0 = 0x80;	[TokenSfr,TokenIdentifier "P0",TokenAssign,	[TokenSfr,TokenIdentifier "P0",TokenAssign,

35	sfr16 TMR0 = 0x8C; void stub(void) { return; }	TokenNumber 128,TokenSemicolon,TokenSfr16, TokenIdentifier "TMR0",TokenAssign,TokenNumber 140, TokenSemicolon,TokenVoid,TokenIdentifier "stub", TokenLeftParen,TokenVoid,TokenRightParen,TokenLeftBrace, TokenReturn,TokenSemicolon,TokenRightBrace]	TokenNumber 128,TokenSemicolon,TokenSfr16, TokenIdentifier "TMR0",TokenAssign,TokenNumber 140, TokenSemicolon,TokenVoid,TokenIdentifier "stub", TokenLeftParen,TokenVoid,TokenRightParen,TokenLeftBrace, TokenReturn,TokenSemicolon,TokenRightBrace]
36	data unsigned char dvar; idata int ivar; xdata unsigned char xbuf; code unsigned char cbyte = 0; void main(void) { return; }	[TokenData,TokenUnsigned,TokenChar, TokenIdentifier "dvar",TokenSemicolon,TokenIdata, TokenInt,TokenIdentifier "ivar",TokenSemicolon, TokenXdata,TokenUnsigned,TokenChar, TokenIdentifier "xbuf",TokenSemicolon,TokenCode, TokenUnsigned,TokenChar,TokenIdentifier "cbyte", TokenAssign,TokenNumber 0,TokenSemicolon,TokenVoid, TokenIdentifier "main",TokenLeftParen,TokenVoid, TokenRightParen,TokenLeftBrace,TokenReturn, TokenSemicolon,TokenRightBrace]	[TokenData,TokenUnsigned,TokenChar, TokenIdentifier "dvar",TokenSemicolon,TokenIdata, TokenInt,TokenIdentifier "ivar",TokenSemicolon, TokenXdata,TokenUnsigned,TokenChar, TokenIdentifier "xbuf",TokenSemicolon,TokenCode, TokenUnsigned,TokenChar,TokenIdentifier "cbyte", TokenAssign,TokenNumber 0,TokenSemicolon,TokenVoid, TokenIdentifier "main",TokenLeftParen,TokenVoid, TokenRightParen,TokenLeftBrace,TokenReturn, TokenSemicolon,TokenRightBrace]
37	void uart_isr(void) interrupt 4 using 1 { return; }	[TokenVoid,TokenIdentifier "uart_isr",TokenLeftParen, TokenVoid,TokenRightParen,TokenInterrupt,TokenNumber 4, TokenUsing,TokenNumber 1,TokenLeftBrace,TokenReturn, TokenSemicolon,TokenRightBrace]	[TokenVoid,TokenIdentifier "uart_isr",TokenLeftParen, TokenVoid,TokenRightParen,TokenInterrupt,TokenNumber 4, TokenUsing,TokenNumber 1,TokenLeftBrace,TokenReturn, TokenSemicolon,TokenRightBrace]
38	bit ready_flag; void main(void) { return; }	[TokenBit,TokenIdentifier "ready_flag",TokenSemicolon, TokenVoid,TokenIdentifier "main",TokenLeftParen, TokenVoid,TokenRightParen,TokenLeftBrace,TokenReturn, TokenSemicolon,TokenRightBrace]	[TokenBit,TokenIdentifier "ready_flag",TokenSemicolon, TokenVoid,TokenIdentifier "main",TokenLeftParen, TokenVoid,TokenRightParen,TokenLeftBrace,TokenReturn, TokenSemicolon,TokenRightBrace]
39	sbit LED_PIN = P1 ^ 1; void main(void) { return; }	[TokenSbit,TokenIdentifier "LED_PIN",TokenAssign, TokenIdentifier "P1",TokenCaret,TokenNumber 1, TokenSemicolon,TokenVoid,TokenIdentifier "main", TokenLeftParen,TokenVoid,TokenRightParen,TokenLeftBrace, TokenReturn,TokenSemicolon,TokenRightBrace]	[TokenSbit,TokenIdentifier "LED_PIN",TokenAssign, TokenIdentifier "P1",TokenCaret,TokenNumber 1, TokenSemicolon,TokenVoid,TokenIdentifier "main", TokenLeftParen,TokenVoid,TokenRightParen,TokenLeftBrace, TokenReturn,TokenSemicolon,TokenRightBrace]
40	reentrant void helper(void) { return; }	[TokenReentrant,TokenVoid,TokenIdentifier "helper", TokenLeftParen,TokenVoid,TokenRightParen,TokenLeftBrace, TokenReturn,TokenSemicolon,TokenRightBrace]	[TokenReentrant,TokenVoid,TokenIdentifier "helper", TokenLeftParen,TokenVoid,TokenRightParen,TokenLeftBrace, TokenReturn,TokenSemicolon,TokenRightBrace]
41	int main() { return 0; }	[TokenInt,TokenIdentifier "main",TokenLeftParen, TokenRightParen,TokenLeftBrace,TokenReturn,TokenNumber 0,	[TokenInt,TokenIdentifier "main",TokenLeftParen, TokenRightParen,TokenLeftBrace,TokenReturn,TokenNumber 0,

		TokenSemicolon,TokenRightBrace]	TokenSemicolon,TokenRightBrace]
42	int main() { return 0; }	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstReturn (Just (ExprLitInt 0))])]	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstReturn (Just (ExprLitInt 0))])]
43	int main() { if (a) return 1; else return 2; }	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstIf (ExprVar "a") (AstCompound [AstReturn (Just (ExprLitInt 1)), AstReturn (Just (ExprLitInt 2))])])]	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstIf (ExprVar "a") (AstCompound [AstReturn (Just (ExprLitInt 1)), AstReturn (Just (ExprLitInt 2))])])]
44	int main() { for (;;) ; }	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstFor Nothing Nothing Nothing (AstExprStmt Nothing)])]	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstFor Nothing Nothing Nothing (AstExprStmt Nothing)])]
45	int main() { while (a) return 1; }	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstWhile (ExprVar "a") (AstReturn (Just (ExprLitInt 1))])])]	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstWhile (ExprVar "a") (AstReturn (Just (ExprLitInt 1))])])]
46	int main() { { return 3; } }	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstCompound [AstReturn (Just (ExprLitInt 3))])])]	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstCompound [AstReturn (Just (ExprLitInt 3))])])]
47	int main() { a = 1; }	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstExprStmt (Just (ExprAssign AAssign (ExprVar "a") (ExprLitInt 1))])])]	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstExprStmt (Just (ExprAssign AAssign (ExprVar "a") (ExprLitInt 1))])])]
48	void timer0(void) interrupt 1 using 2 { return; }	AstProgram [AstFunctionDef "timer0" [TokenVoid] [TokenInterrupt, TokenNumber 1,TokenUsing, TokenNumber 2] (AstCompound [AstReturn Nothing])]	AstProgram [AstFunctionDef "timer0" [TokenVoid] [TokenInterrupt, TokenNumber 1,TokenUsing, TokenNumber 2] (AstCompound [AstReturn Nothing])]
49	int main(void) { switch (x) { case 0: break; } return 0; }	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstSwitch (ExprVar "x") (AstCompound [AstCase (ExprLitInt 0) AstBreak]), AstReturn (Just (ExprLitInt 0))])]	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstSwitch (ExprVar "x") (AstCompound [AstCase (ExprLitInt 0) AstBreak]), AstReturn (Just (ExprLitInt 0))])]
50	int main(void) { do { x = 1; } while (x);	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstDoWhile (AstCompound [AstExprStmt (Just (ExprAssign AAssign (ExprVar "x") (ExprLitInt 1))]) (ExprVar "x"), AstReturn (Just (ExprLitInt 0))])])]	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstDoWhile (AstCompound [AstExprStmt (Just (ExprAssign AAssign (ExprVar "x") (ExprLitInt 1))]) (ExprVar "x"), AstReturn (Just (ExprLitInt 0))])])]

	return 0; }		
51	struct S { int x; }; int main(void) { return 0; }	AstProgram [AstDeclaration [TokenStruct, TokenIdentifier "S",TokenLeftBrace,TokenInt, TokenIdentifier "x",TokenSemicolon,TokenRightBrace, TokenSemicolon], AstFunctionDef "main" [TokenInt] [] (AstCompound [AstReturn (Just (ExprLitInt 0))])]	AstProgram [AstDeclaration [TokenStruct, TokenIdentifier "S",TokenLeftBrace,TokenInt, TokenIdentifier "x",TokenSemicolon,TokenRightBrace, TokenSemicolon], AstFunctionDef "main" [TokenInt] [] (AstCompound [AstReturn (Just (ExprLitInt 0))])]
52	int main(void) { switch (y) { default: break; } return 1; }	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstSwitch (ExprVar "y") (AstCompound [AstDefault AstBreak]), AstReturn (Just (ExprLitInt 1))])]	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstSwitch (ExprVar "y") (AstCompound [AstDefault AstBreak]), AstReturn (Just (ExprLitInt 1))])]
53	int main(void) { switch (v) { case 0: case 1: x = 2; break; } return 0; }	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstSwitch (ExprVar "v") (AstCompound [AstCase (ExprLitInt 0) (AstCase (ExprLitInt 1) (AstExprStmt (Just (ExprAssign AAssign (ExprVar "x") (ExprLitInt 2)))]), AstBreak]),AstReturn (Just (ExprLitInt 0))])]	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstSwitch (ExprVar "v") (AstCompound [AstCase (ExprLitInt 0) (AstCase (ExprLitInt 1) (AstExprStmt (Just (ExprAssign AAssign (ExprVar "x") (ExprLitInt 2)))]), AstBreak]),AstReturn (Just (ExprLitInt 0))])]
54	int main() { return 200; }	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstReturn (Just (ExprLitInt 200))])]	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstReturn (Just (ExprLitInt 200))])]
55	sfr P0 = 0x80; sfr16 TMR0 = 0x8C; void stub(void) { return; }	AstProgram [AstDeclaration [TokenSfr, TokenIdentifier "P0",TokenAssign,TokenNumber 128, TokenSemicolon],AstDeclaration [TokenSfr16, TokenIdentifier "TMR0",TokenAssign,TokenNumber 140, TokenSemicolon],	AstProgram [AstDeclaration [TokenSfr, TokenIdentifier "P0",TokenAssign,TokenNumber 128, TokenSemicolon],AstDeclaration [TokenSfr16, TokenIdentifier "TMR0",TokenAssign,TokenNumber 140, TokenSemicolon],

	}	AstFunctionDef "stub" [TokenVoid] [] (AstCompound [AstReturn Nothing])]	AstFunctionDef "stub" [TokenVoid] [] (AstCompound [AstReturn Nothing])]
56	data unsigned char dvar; idata int ivar; xdata unsigned char xbuf; code unsigned char cbyte = 0; void main(void) { return; }	AstProgram [AstDeclaration [TokenData,TokenUnsigned, TokenChar,TokenIdentifier "dvar",TokenSemicolon], AstDeclaration [TokenIddata,TokenInt, TokenIdentifier "ivar",TokenSemicolon], AstDeclaration [TokenXdata,TokenUnsigned,TokenChar, TokenIdentifier "xbuf",TokenSemicolon], AstDeclaration [TokenCode,TokenUnsigned,TokenChar, TokenIdentifier "cbyte",TokenAssign,TokenNumber 0, TokenSemicolon], AstFunctionDef "main" [TokenVoid] [] (AstCompound [AstReturn Nothing])]	AstProgram [AstDeclaration [TokenData,TokenUnsigned, TokenChar,TokenIdentifier "dvar",TokenSemicolon], AstDeclaration [TokenIddata,TokenInt, TokenIdentifier "ivar",TokenSemicolon], AstDeclaration [TokenXdata,TokenUnsigned,TokenChar, TokenIdentifier "xbuf",TokenSemicolon], AstDeclaration [TokenCode,TokenUnsigned,TokenChar, TokenIdentifier "cbyte",TokenAssign,TokenNumber 0, TokenSemicolon], AstFunctionDef "main" [TokenVoid] [] (AstCompound [AstReturn Nothing])]
57	void uart_isr(void) interrupt 4 using 1 { return; }	AstProgram [AstFunctionDef "uart_isr" [TokenVoid] [TokenInterrupt, TokenNumber 4,TokenUsing, TokenNumber 1] (AstCompound [AstReturn Nothing])]	AstProgram [AstFunctionDef "uart_isr" [TokenVoid] [TokenInterrupt, TokenNumber 4,TokenUsing, TokenNumber 1] (AstCompound [AstReturn Nothing])]
58	bit ready_flag; void main(void) { return; }	AstProgram [AstDeclaration [TokenBit, TokenIdentifier "ready_flag",TokenSemicolon], AstFunctionDef "main" [TokenVoid] [] (AstCompound [AstReturn Nothing])]	AstProgram [AstDeclaration [TokenBit, TokenIdentifier "ready_flag",TokenSemicolon], AstFunctionDef "main" [TokenVoid] [] (AstCompound [AstReturn Nothing])]
59	sbit LED_PIN = P1 ^ 1; void main(void) { return; }	AstProgram [AstDeclaration [TokenSbit, TokenIdentifier "LED_PIN",TokenAssign, TokenIdentifier "P1",TokenCaret,TokenNumber 1, TokenSemicolon], AstFunctionDef "main" [TokenVoid] [] (AstCompound [AstReturn Nothing])]	AstProgram [AstDeclaration [TokenSbit, TokenIdentifier "LED_PIN",TokenAssign, TokenIdentifier "P1",TokenCaret,TokenNumber 1, TokenSemicolon], AstFunctionDef "main" [TokenVoid] [] (AstCompound [AstReturn Nothing])]
60	reentrant void helper(void) { return; }	AstProgram [AstFunctionDef "helper" [TokenReentrant, TokenVoid] [] (AstCompound [AstReturn Nothing])]	AstProgram [AstFunctionDef "helper" [TokenReentrant, TokenVoid] [] (AstCompound [AstReturn Nothing])]
61	int main() { return 0; }	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstReturn (Just (ExprLitInt 0))])]	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstReturn (Just (ExprLitInt 0))])]
62	int main(){ return a+b*c; }	ExprBinary OpAdd (ExprVar "a") (ExprBinary OpMul (ExprVar "b") (ExprVar "c"))	ExprBinary OpAdd (ExprVar "a") (ExprBinary OpMul (ExprVar "b") (ExprVar "c"))
63	int main(){ return a*b+c; }	ExprBinary OpAdd (ExprBinary OpMul (ExprVar "a") (ExprVar "b")) (ExprVar "c")	ExprBinary OpAdd (ExprBinary OpMul (ExprVar "a") (ExprVar "b")) (ExprVar "c")

64	int main(){ return a-b-c; }	ExprBinary OpSub (ExprBinary OpSub (ExprVar "a") (ExprVar "b")) (ExprVar "c")	ExprBinary OpSub (ExprBinary OpSub (ExprVar "a") (ExprVar "b")) (ExprVar "c")
65	int main(){ return a<b+c; }	ExprBinary OpShl (ExprVar "a") (ExprBinary OpAdd (ExprVar "b") (ExprVar "c"))	ExprBinary OpShl (ExprVar "a") (ExprBinary OpAdd (ExprVar "b") (ExprVar "c"))
66	int main(){ return a+b==c; }	ExprBinary OpEq (ExprBinary OpAdd (ExprVar "a") (ExprVar "b")) (ExprVar "c")	ExprBinary OpEq (ExprBinary OpAdd (ExprVar "a") (ExprVar "b")) (ExprVar "c")
67	int main(){ return a b&c; }	ExprBinary OpBitOr (ExprVar "a") (ExprBinary OpBitAnd (ExprVar "b") (ExprVar "c"))	ExprBinary OpBitOr (ExprVar "a") (ExprBinary OpBitAnd (ExprVar "b") (ExprVar "c"))
68	int main(){ return a^b c; }	ExprBinary OpBitOr (ExprBinary OpBitXor (ExprVar "a") (ExprVar "b")) (ExprVar "c")	ExprBinary OpBitOr (ExprBinary OpBitXor (ExprVar "a") (ExprVar "b")) (ExprVar "c")
69	int main(){ return a b&& c; }	ExprBinary OpOr (ExprVar "a") (ExprBinary OpAnd (ExprVar "b") (ExprVar "c"))	ExprBinary OpOr (ExprVar "a") (ExprBinary OpAnd (ExprVar "b") (ExprVar "c"))
70	int main(){ return (a+b)*c; }	ExprBinary OpMul (ExprBinary OpAdd (ExprVar "a") (ExprVar "b")) (ExprVar "c")	ExprBinary OpMul (ExprBinary OpAdd (ExprVar "a") (ExprVar "b")) (ExprVar "c")
71	int main(){ return a?b:c?d:e; }	ExprTernary (ExprVar "a") (ExprVar "b") (ExprTernary (ExprVar "c") (ExprVar "d") (ExprVar "e"))	ExprTernary (ExprVar "a") (ExprVar "b") (ExprTernary (ExprVar "c") (ExprVar "d") (ExprVar "e"))
72	int main(){ return a=b=c; }	ExprAssign AAssign (ExprVar "a") (ExprAssign AAssign (ExprVar "b") (ExprVar "c"))	ExprAssign AAssign (ExprVar "a") (ExprAssign AAssign (ExprVar "b") (ExprVar "c"))
73	int main(){ return a<b==c; }	ExprBinary OpEq (ExprBinary OpLt (ExprVar "a") (ExprVar "b")) (ExprVar "c")	ExprBinary OpEq (ExprBinary OpLt (ExprVar "a") (ExprVar "b")) (ExprVar "c")
74	[TokenInt,TokenIdentifier "x"]	AstUnknown [TokenInt,TokenIdentifier "x"]	AstUnknown [TokenInt,TokenIdentifier "x"]
75	int main() { return 0; }	int main() { return 0; }	int main() { return 0; }
76	a b c	a b c	a b c
77	??=define A 1 int x ??(0 ??) ;	int x [0] ;	int x [0] ;
78	#define N 42 int y = N;	int y = 42;	int y = 42;
79	#define FOO #ifdef FOO int a = 1; #else	int a = 1;	int a = 1;

	int a = 2; #endif		
80	#ifdef FOO int b = 1; #else int b = 2; #endif	int b = 2;	int b = 2;
81	#ifndef H_H #define H_H int z = 1; #endif #ifndef H_H int z = 2; #endif	int z = 1;	int z = 1;
82	#include <angle.h> #include "quoted.h" int v = ANGLE + QUOTED;	int v = 40 + 2;	int v = 40 + 2;
83	int main() { return 7; }	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstReturn (Just (ExprLitInt 7))])]	AstProgram [AstFunctionDef "main" [TokenInt] [] (AstCompound [AstReturn (Just (ExprLitInt 7))])]
84	[TokenInt,TokenIdentifier "x"]	AstUnknown [TokenInt,TokenIdentifier "x"]	AstUnknown [TokenInt,TokenIdentifier "x"]
85	int x;	AstProgram [AstDeclaration [TokenInt,TokenIdentifier "x", TokenSemicolon]]	AstProgram [AstDeclaration [TokenInt,TokenIdentifier "x", TokenSemicolon]]
86	[TokenInt,TokenIdentifier "x",TokenSemicolon]	Decl {declSpecifiers = [SpecType TyInt], declDeclarators = [Declarator {dtorName = "x", dtorSuffixes = [], dtorInit = Nothing}]}	Decl {declSpecifiers = [SpecType TyInt], declDeclarators = [Declarator {dtorName = "x", dtorSuffixes = [], dtorInit = Nothing}]}
87	[TokenUnsigned,Token Char,TokenIdentifier "buf", TokenLeftBracket,Token Number 10,TokenRightBracket, TokenSemicolon]	Decl {declSpecifiers = [SpecType TyUnsigned, SpecType TyChar], declDeclarators = [Declarator {dtorName = "buf",	Decl {declSpecifiers = [SpecType TyUnsigned, SpecType TyChar], declDeclarators = [Declarator {dtorName = "buf",

		dtorSuffixes = [DtorArray (Just (ELitInt 10))), dtorInit = Nothing]]}	dtorSuffixes = [DtorArray (Just (ELitInt 10))), dtorInit = Nothing]]}
88	[TokenData,TokenInt,TokenIdentifier "cnt", TokenSemicolon]	Decl {declSpecifiers = [SpecC51Memory MemData, SpecType TyInt], declDeclarators = [Declarator {dtorName = "cnt", dtorSuffixes = [], dtorInit = Nothing}]}	Decl {declSpecifiers = [SpecC51Memory MemData, SpecType TyInt], declDeclarators = [Declarator {dtorName = "cnt", dtorSuffixes = [], dtorInit = Nothing}]}
89	[TokenUnsigned,TokenChar,TokenIdentifier "x",TokenAssign, TokenNumber 255,TokenSemicolon]	Decl {declSpecifiers = [SpecType TyUnsigned, SpecType TyChar], declDeclarators = [Declarator {dtorName = "x", dtorSuffixes = [], dtorInit = Just (ELitInt 255)}]}	Decl {declSpecifiers = [SpecType TyUnsigned, SpecType TyChar], declDeclarators = [Declarator {dtorName = "x", dtorSuffixes = [], dtorInit = Just (ELitInt 255)}]}
90	[TokenInt,TokenIdentifier "a",TokenComma, TokenIdentifier "b",TokenSemicolon]	Decl {declSpecifiers = [SpecType TyInt], declDeclarators = [Declarator {dtorName = "a", dtorSuffixes = [], dtorInit = Nothing}, Declarator {dtorName = "b", dtorSuffixes = [], dtorInit = Nothing}]}	Decl {declSpecifiers = [SpecType TyInt], declDeclarators = [Declarator {dtorName = "a", dtorSuffixes = [], dtorInit = Nothing}, Declarator {dtorName = "b", dtorSuffixes = [], dtorInit = Nothing}]}
91	[TokenSfr,TokenIdentifier "P0",TokenAssign, TokenNumber 128,TokenSemicolon]	Decl {declSpecifiers = [SpecType TySfr], declDeclarators = [Declarator {dtorName = "P0", dtorSuffixes = [], dtorInit = Just (ELitInt 128)}]}	Decl {declSpecifiers = [SpecType TySfr], declDeclarators = [Declarator {dtorName = "P0", dtorSuffixes = [], dtorInit = Just (ELitInt 128)}]}
92	int main() { return 7; }	ASTProgram (Program {programDecls = [ExtFunction (FunctionDef {fnName = "main", fnDeclSpecs = [SpecType TyInt], fnC51AttrTokens = [], fnBody = SCompound [SReturn (Just (ELitInt 7))]}])])	ASTProgram (Program {programDecls = [ExtFunction (FunctionDef {fnName = "main", fnDeclSpecs = [SpecType TyInt], fnC51AttrTokens = [], fnBody = SCompound [SReturn (Just (ELitInt 7))]}])])
93	int x;	ASTProgram (Program {programDecls = [ExtDecl (Decl {declSpecifiers = [SpecType TyInt], declDeclarators = [Declarator {dtorName = "x", dtorSuffixes = [], dtorInit = Nothing}]})])])	ASTProgram (Program {programDecls = [ExtDecl (Decl {declSpecifiers = [SpecType TyInt], declDeclarators = [Declarator {dtorName = "x", dtorSuffixes = [], dtorInit = Nothing}]})])])
	struct S { int x; };	length decls == 2; ExtDeclUnparsed : ExtFunction	ASTProgram (Program {programDecls = [ExtDeclUnparsed [TokenStruct,

